1 Vector calculus in numerator layout

We use $\partial x_i/\partial x_j = \delta_{ij}$ and column vectors throughout. For scalar f,

$$\frac{\partial f}{\partial \mathbf{x}} \in \mathbb{R}^{1\times n}, \qquad \nabla_{\mathbf{x}} f = \left(\frac{\partial f}{\partial \mathbf{x}}\right)^{\mathsf{T}} \in \mathbb{R}^{n\times 1}.$$

(a) Linear maps.

$$\frac{\partial (A\mathbf{x})_i}{\partial x_j} = \sum_{\ell=1}^{n} A_{i\ell}\delta_{\ell j} = A_{ij}, \qquad \frac{\partial (A\mathbf{x})}{\partial \mathbf{x}} = A \in \mathbb{R}^{m\times n},$$

$$\frac{\partial (\mathbf{c}^{\mathsf{T}}\mathbf{x})}{\partial x_j} = \sum_{i=1}^{n} c_i \delta_{ij} = c_j, \qquad \frac{\partial (\mathbf{c}^{\mathsf{T}}\mathbf{x})}{\partial \mathbf{x}} = \mathbf{c}^{\mathsf{T}} \in \mathbb{R}^{1\times n}.$$

Hence $\nabla_{\mathbf{x}}(\mathbf{c}^{\mathsf{T}}\mathbf{x}) = \mathbf{c} \in \mathbb{R}^{n\times 1}$.

(b) Product rule.

$$\frac{\partial}{\partial x_j}\sum_{i=1}^{k} a_i(\mathbf{x}) b_i(\mathbf{x}) = \sum_{i=1}^{k}\left(a_i \frac{\partial b_i}{\partial x_j} + b_i \frac{\partial a_i}{\partial x_j}\right).$$

Thus

$$\underbrace{\frac{\partial (\mathbf{a}^{\mathsf{T}}\mathbf{b})}{\partial \mathbf{x}}}_{1\times n} = \underbrace{\mathbf{a}^{\mathsf{T}}}_{1\times k}\underbrace{\frac{\partial \mathbf{b}}{\partial \mathbf{x}}}_{k\times n} + \underbrace{\mathbf{b}^{\mathsf{T}}}_{1\times k}\underbrace{\frac{\partial \mathbf{a}}{\partial \mathbf{x}}}_{k\times n}.$$

(c) Quadratic forms. With $\mathbf{a} = \mathbf{x}$ and $\mathbf{b} = B\mathbf{x}$,

$$\frac{\partial q}{\partial \mathbf{x}} = \mathbf{x}^{\mathsf{T}}B + (B\mathbf{x})^{\mathsf{T}} = \mathbf{x}^{\mathsf{T}}(B + B^{\mathsf{T}}) \in \mathbb{R}^{1\times n}.$$

In indices, the two terms are

$$\frac{\partial}{\partial x_j}\sum_{i,\ell=1}^{n} x_i B_{i\ell} x_\ell = \sum_{\ell=1}^{n} B_{j\ell} x_\ell + \sum_{i=1}^{n} x_i B_{ij}.$$

Therefore

$$\nabla_{\mathbf{x}} q = (B + B^{\mathsf{T}})\mathbf{x} \in \mathbb{R}^{n\times 1}.$$

The special cases are

	$\partial q/\partial \mathbf{x}$	$\nabla_{\mathbf{x}} q$
$B = B^{\mathsf{T}}$	$2\mathbf{x}^{\mathsf{T}}B$	$2B\mathbf{x}$
$B = I$	$2\mathbf{x}^{\mathsf{T}}$	$2\mathbf{x}$

(d) Chain rule and a squared residual. With $\mathbf{y} = \mathbf{g}(\mathbf{x})$, the scalar chain rule gives

$$\frac{\partial h_i(\mathbf{g}(\mathbf{x}))}{\partial x_j} = \sum_{\ell=1}^{m} \left.\frac{\partial h_i}{\partial y_\ell}\right|_{\mathbf{y}=\mathbf{g}(\mathbf{x})} \frac{\partial g_\ell}{\partial x_j}.$$

This is matrix multiplication:

$$\underbrace{\frac{\partial (\mathbf{h}\circ\mathbf{g})}{\partial \mathbf{x}}}_{r\times n} = \underbrace{\left.\frac{\partial \mathbf{h}}{\partial \mathbf{y}}\right|_{\mathbf{y}=\mathbf{g}(\mathbf{x})}}_{r\times m}\underbrace{\frac{\partial \mathbf{g}}{\partial \mathbf{x}}}_{m\times n}.$$

For $\mathbf{y} = A\mathbf{x} - \mathbf{d}$ and $L = \mathbf{y}^\mathsf{T}\mathbf{y}$, the two chain-rule factors are

$$\frac{\partial(\mathbf{y}^\mathsf{T}\mathbf{y})}{\partial \mathbf{y}} = 2\mathbf{y}^\mathsf{T} \in \mathbb{R}^{1\times m}, \qquad \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = A \in \mathbb{R}^{m\times n}.$$

Hence

$$\begin{aligned}
\frac{\partial L}{\partial x_j} &= 2\sum_{i=1}^{m} y_i A_{ij}, \\
\frac{\partial L}{\partial \mathbf{x}} &= \underbrace{2\mathbf{y}^\mathsf{T}}_{1\times m}\underbrace{A}_{m\times n} = 2(A\mathbf{x}-\mathbf{d})^\mathsf{T}A \in \mathbb{R}^{1\times n}, \\
\nabla_{\mathbf{x}} L &= 2A^\mathsf{T}(A\mathbf{x}-\mathbf{d}) \in \mathbb{R}^{n\times 1}.
\end{aligned}$$

Check by expansion:

$$\begin{aligned}
L &= \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x} - 2\mathbf{d}^\mathsf{T}A\mathbf{x} + \mathbf{d}^\mathsf{T}\mathbf{d}, \\
\nabla_{\mathbf{x}} L &= 2A^\mathsf{T}A\mathbf{x} - 2A^\mathsf{T}\mathbf{d} = 2A^\mathsf{T}(A\mathbf{x}-\mathbf{d}).
\end{aligned}$$

2 PCA as non-convex optimization problem

Write $C\mathbf{e}_i = \lambda_i \mathbf{e}_i$ for an orthonormal eigenbasis, with $\lambda_1 \geq \lambda_2 \geq 0$. The initial population covariance $\Sigma_{1/2}$ has eigenvalues $3/2$ and $1/2$ and PC1 $\pm(1,1)^\mathsf{T}/\sqrt{2}$; finite-sample values generally differ.

(b) Parameter effects. In eigenvector coordinates,

$$-V(\mathbf{w}) = -\lambda_1(\mathbf{e}_1^\mathsf{T}\mathbf{w})^2 - \lambda_2(\mathbf{e}_2^\mathsf{T}\mathbf{w})^2.$$

For $C \succ 0$, contours are ellipses along the eigenvectors, narrower in directions of larger eigenvalues. On the unit circle, minima are $\pm\mathbf{e}_1$; if $\lambda_1 = \lambda_2$, all directions are optimal.

Feature scaling. For $S = \mathrm{diag}(s_x, s_y)$,

$$C' = SCS^\mathsf{T} = \begin{pmatrix} s_x^2 c_{11} & s_x s_y c_{12} \\ s_x s_y c_{12} & s_y^2 c_{22} \end{pmatrix}.$$

Unequal positive scaling changes the landscape and, for $c_{12} \neq 0$, rotates PC1 toward the increasingly stretched feature. If $c_{12} = 0$, PC1 switches axes when the scaled variances cross. Thus PCA depends on feature units. Equal scaling gives $C' = s^2 C$: directions and contour axis ratios stay fixed, while eigenvalues and objective values scale by s^2.

Correlation. Σ_ρ has eigenvalues $1 \pm |\rho|$. PC1 follows $x_2 = x_1$ for $\rho > 0$ and $x_2 = -x_1$ for $\rho < 0$. Larger $|\rho|$ increases the eigenvalue gap and contour anisotropy; at $\rho = 0$, no population direction is preferred.

Sample count. Increasing n reduces fluctuations in C around the population covariance. PC1 is especially sensitive when the eigenvalue gap is small; for an isotropic population, there is no unique PC1 to recover.

(c) Convergence of normalized gradient ascent. Put $A = I + 2\eta C$ and $\mathbf{w}_0 = a_1\mathbf{e}_1 + a_2\mathbf{e}_2$. Normalization changes only the length, so

$$\mathbf{w}_k = \frac{A^k \mathbf{w}_0}{\|A^k \mathbf{w}_0\|_2}, \qquad A^k \mathbf{w}_0 = \sum_{i=1}^{2} a_i (1 + 2\eta\lambda_i)^k \mathbf{e}_i.$$

For $\lambda_1 > \lambda_2$ and $a_1 \neq 0$, the relative second component decays by

$$\frac{1 + 2\eta\lambda_2}{1 + 2\eta\lambda_1} = \frac{\lambda_1 + 2\lambda_2}{2\lambda_1 + \lambda_2} < 1 \qquad \left(\eta = \frac{1}{2\,\mathrm{tr}\, C}\right),$$

hence $\mathbf{w}_k \to \mathrm{sign}(a_1)\mathbf{e}_1$. However, $\mathbf{w}_0 = \pm\mathbf{e}_2$ stays fixed and is suboptimal: convergence does not hold for every start. In the app, slider precision and rounding introduce a PC1 component, causing eventual escape from this unstable fixed point; a nearby start can linger for many iterations. If $\lambda_1 = \lambda_2$, every unit start is optimal.

(d) Relaxation to the unit disk. Since

$$\mathbf{z}^\mathsf{T} C \mathbf{z} = \frac{1}{n}\|X^\mathsf{T}\mathbf{z}\|_2^2 \geq 0, \qquad \nabla^2 V = 2C \succeq 0,$$

V is convex. The disk is convex by the triangle inequality. For $\mathbf{w} = r\mathbf{q}$, $\|\mathbf{q}\|_2 = 1$, $0 < r \leq 1$,

$$V(\mathbf{w}) = r^2 V(\mathbf{q}) \leq r^2 \lambda_1 \leq \lambda_1.$$

With $\lambda_1 > 0$, equality requires $r = 1$ and $\mathbf{q}$ in the leading eigenspace; $V(\mathbf{0}) = 0$ is suboptimal. Circle and disk therefore have the same maximum and maximizers. The problem is still not convex optimization: we maximize a convex, rather than concave, objective over the convex disk.

(e) Circle and disk trajectories. Write $\mathbf{v}_k = r_k \mathbf{q}_k$ with $r_k > 0$ and $\|\mathbf{q}_k\|_2 = 1$. With $A = I + 2\eta C$, the step before projection is $r_k A \mathbf{q}_k$, giving

$$r_{k+1} = \min\{1, r_k \|A\mathbf{q}_k\|_2\}, \qquad \mathbf{q}_{k+1} = \frac{A\mathbf{q}_k}{\|A\mathbf{q}_k\|_2}.$$

The direction update equals the circle update, so shared initial directions give $\mathbf{q}_k = \mathbf{w}_k$ at every step. Until the disk reaches the boundary, $\mathbf{v}_k = r_0 A^k \mathbf{q}$. Positive definiteness implies

$$r_0 \|A^k \mathbf{q}\|_2 \geq r_0 (1 + 2\eta\lambda_2)^k \longrightarrow \infty,$$

so the first boundary step is finite:

$$K = \min\{k \geq 1 : \ r_0 \|A^k \mathbf{q}\|_2 \geq 1\} \leq \left\lceil \frac{\ln(1/r_0)}{\ln(1 + 2\eta\lambda_2)} \right\rceil.$$

At K, projection gives $\mathbf{v}_K = \mathbf{w}_K$. Since $\|A\mathbf{z}\|_2 \geq 1 + 2\eta\lambda_2 > 1$ for every unit $\mathbf{z}$, both updates agree thereafter: $\mathbf{v}_k = \mathbf{w}_k$ for all $k \geq K$. For $\mathbf{v}_0 = \mathbf{0}$, all iterates remain zero and are suboptimal.

3 RANSAC

Each draw is an inlier with probability p. Since the m points in a sample are drawn independently with replacement, the probability that one sample is outlier-free is p^m. Thus a sample contains at least one outlier with probability $1 - p^m$.

The r samples are independent, so the probability that all of them contain outliers is $(1 - p^m)^r$. Taking the complementary event gives

$$\Pr(\text{at least one outlier-free sample}) = 1 - (1 - p^m)^r.$$

We require this probability to be at least 0.99, hence

$$\begin{aligned} 1 - (1 - p^m)^r &\geq 0.99, \\ (1 - p^m)^r &\leq 0.01, \\ r \ln(1 - p^m) &\leq \ln(0.01). \end{aligned}$$

Since $0 < p < 1$ and $m \geq 1$, we have $\ln(1 - p^m) < 0$. Dividing by this negative number reverses the inequality:

$$r \geq \frac{\ln(0.01)}{\ln(1 - p^m)}.$$

Therefore the smallest integer number of repeats that achieves the required probability is

$$\boxed{r_{\min} = \left\lceil \frac{\ln(0.01)}{\ln(1 - p^m)} \right\rceil}.$$